Ubuntu Terminal Mastery Guide

System Info • Disk • RAM • Aesthetics • Terminal Customization

Complete Reference with Commands & Syntax

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1. System Information Commands

These commands give you a full overview of your system directly from the terminal. No GUI needed.

1.1 All-in-One Summary — neofetch

neofetch is the most popular tool for displaying system info with a distro logo in ASCII art.

Install neofetch:

```
sudo apt install neofetch
```

Run it:

```
neofetch
```

Sample output info shown by neofetch includes: OS, Host, Kernel, Uptime, Packages, Shell, Resolution, DE, WM, Theme, Icons, Terminal, CPU, GPU, and Memory.

> **NOTE:** Your system shows: Intel i7-8700K (12) @ 4.700GHz, NVIDIA GeForce GTX 1080 Ti, 48GB RAM on Ubuntu 24.04.4 LTS.

1.2 Custom ASCII Distro Logos

You can change the ASCII art logo shown with neofetch:

```
neofetch --ascii_distro ubuntu_small
neofetch --ascii_distro arch
neofetch --ascii_distro windows
neofetch --ascii_distro debian
neofetch --ascii_distro kali
```

2. Disk Space & Storage

2.1 df — Disk Free

Shows used and available space across all mounted filesystems.

Basic usage:

```
df -h
```

Check only the root / partition:

```
df -h /
```

Item	Description
`df -h`	Human-readable sizes (GB/MB)
`df -H`	SI units (1000 not 1024)
`df -T`	Show filesystem type
`df -i`	Show inode usage instead of blocks

Your output explained:

Item	Description
`/dev/nvme0n1p2`	Your 256GB NVMe SSD — main system drive
`233G total`	Usable size after formatting overhead
`19G used`	Currently only 19GB used (9%)
`203G available`	Plenty of free space remaining

2.2 lsblk — List Block Devices

Shows all disks and partitions in a tree layout.

```
lsblk
```

With more detail:

```
lsblk -o NAME,SIZE,TYPE,MOUNTPOINT
```

Detect SSD vs HDD:

```
lsblk -d -o NAME,SIZE,ROTA
```

Item	Description
`ROTA = 0`	SSD (Solid State Drive) — no rotation
`ROTA = 1`	HDD (Hard Disk Drive) — spinning platter

2.3 ncdu — Visual Disk Usage

An interactive, ncurses-based disk usage viewer. Great for finding what is eating space.

```
sudo apt install ncdu
ncdu /
```

Navigate folders with arrow keys. Press d to delete, q to quit.

2.4 du — Directory Usage

Summarize disk usage of a specific folder:

```
du -h --max-depth=1 /
du -sh /home/yourusername
```

Item	Description
`-s`	Summary only (single total)
`-h`	Human-readable
`--max-depth=1`	Only go 1 folder level deep

3. RAM & Memory

3.1 free — Memory Overview

```
free -h
```

Output columns explained:

Item	Description
`total`	All installed RAM

`used`	Currently in use by OS + apps
`free`	Completely unused RAM
`shared`	Used by shared memory (tmpfs)
`buff/cache`	Used as disk cache (can be freed if needed)
`available`	RAM actually available to new processes

 NOTE: Your system has 48GB RAM. The 'available' column is more useful than 'free' — Linux uses spare RAM as disk cache to speed things up.

3.2 vmstat — Virtual Memory Stats

```
vmstat
vmstat -s
vmstat 1 5   # update every 1s, 5 times
```

4. CPU Information

4.1 lscpu

```
lscpu
```

Shows: Architecture, CPU model, cores, threads, speed, cache sizes, and flags.

4.2 Live CPU Usage

```
watch -n 1 cat /proc/loadavg
```

4.3 CPU Temperature

```
sudo apt install lm-sensors
sensors
```

5. System Monitor Tools

5.1 top — Built-in Process Monitor

```
top
```

Press q to quit. Press M to sort by memory. Press P to sort by CPU.

5.2 htop — Better Interactive Monitor

```
sudo apt install htop
htop
```

htop shows color-coded CPU bars, memory usage, and a process list. Use F-keys to filter, kill, or sort processes.

5.3 btop — Best Animated Dashboard

btop is the most visually impressive terminal monitor. It shows animated CPU, RAM, disk, and network graphs.

```
sudo apt install btop
btop
```

Works on Ubuntu Server too (via SSH from a good terminal).

5.4 GPU Live Stats (NVIDIA)

Since you have a GTX 1080 Ti, this shows live GPU stats:

```
watch -n 1 nvidia-smi
```

Updates every 1 second. Shows VRAM used, GPU temperature, power draw, and running processes.

6. Aesthetic Terminal Tools

6.1 neofetch (Classic)

The original. Shows system info with a distro ASCII logo.

```
sudo apt install neofetch
neofetch
```

6.2 fastfetch (Modern & Faster)

A modern replacement for neofetch. Faster startup, more customizable. Not in Ubuntu 24.04 default repos — install from GitHub:

```
wget -qO fastfetch.tar.gz https://github.com/fastfetch-cli/fastfetch/releases/latest/download/fastfetch-linux-amd64.tar.gz
sudo tar xf fastfetch.tar.gz --strip-components=3 -C /usr/local/bin fastfetch-linux-amd64/usr/bin/fastfetch
rm fastfetch.tar.gz
fastfetch
```

 NOTE: If sudo apt install fastfetch fails (E: Unable to locate package), use the wget method above. Ubuntu 24.04 does not include it in default repos.

6.3 screenfetch

```
sudo apt install screenfetch
screenfetch
```

6.4 pfetch (Minimal)

```
sudo apt install pfetch
pfetch
```

7. Terminal Color Customization

Ubuntu's default GNOME Terminal uses a dark-grey background, not pure black. Here's how to fix it with terminal commands.

Step 1 — Get Your Profile ID

```
gsettings get org.gnome.Terminal.ProfilesList default
```

You will see output like:

```
'b1dcc9dd-5262-4d8d-a863-c897e6d979b9'
```

Copy the value inside the quotes — that is your profile ID.

Step 2 — Apply Full Black Theme

Replace YOUR_PROFILE_ID with your actual ID. Here is the complete one-liner:

```
PROFILE=$(gsettings get org.gnome.Terminal.ProfilesList default | tr -d \');  \
gsettings set
org.gnome.Terminal.Legacy.Profile:/org/gnome/terminal/legacy/profiles:/:$PROFILE/
use-theme-colors false;  \
gsettings set
org.gnome.Terminal.Legacy.Profile:/org/gnome/terminal/legacy/profiles:/:$PROFILE/
background-color 'rgb(0,0,0)';  \
gsettings set
org.gnome.Terminal.Legacy.Profile:/org/gnome/terminal/legacy/profiles:/:$PROFILE/
foreground-color 'rgb(255,255,255)';  \
gsettings set
org.gnome.Terminal.Legacy.Profile:/org/gnome/terminal/legacy/profiles:/:$PROFILE/
use-transparent-background false
```

Step 3 — Your Specific Command (ID already filled in)

Since your profile ID is b1dcc9dd-5262-4d8d-a863-c897e6d979b9, here are the exact commands:

```
gsettings set
org.gnome.Terminal.Legacy.Profile:/org/gnome/terminal/legacy/profiles:/:b1dcc9dd-
5262-4d8d-a863-c897e6d979b9/ use-theme-colors false

gsettings set
org.gnome.Terminal.Legacy.Profile:/org/gnome/terminal/legacy/profiles:/:b1dcc9dd-
5262-4d8d-a863-c897e6d979b9/ background-color 'rgb(0,0,0)'

gsettings set
org.gnome.Terminal.Legacy.Profile:/org/gnome/terminal/legacy/profiles:/:b1dcc9dd-
5262-4d8d-a863-c897e6d979b9/ foreground-color 'rgb(255,255,255)'
```

```
gsettings set
org.gnome.Terminal.Legacy.Profile:/org/gnome/terminal/legacy/profiles:/:b1dcc9dd-
5262-4d8d-a863-c897e6d979b9/ use-transparent-background false
```

Step 4 — Disable Transparency (Extra fix if still grey)

```
gsettings set
org.gnome.Terminal.Legacy.Profile:/org/gnome/terminal/legacy/profiles:/:b1dcc9dd-
5262-4d8d-a863-c897e6d979b9/ background-transparency-percent 0
```

After running these commands, close the terminal and reopen it. The background should now be pure black (#000000).

Custom Colors Reference

Item	Description
`rgb(0,0,0)`	Pure black background
`rgb(255,255,255)`	White text
`rgb(0,255,65)`	Matrix green text
`rgb(80,250,123)`	Dracula green
`rgb(139,233,253)`	Dracula cyan
`rgb(255,121,198)`	Dracula pink

8. Useful Bonus Commands

8.1 System Uptime

```
uptime
```

8.2 Hostname and IP

```
hostname
hostname -I
ip addr show
```

8.3 OS Version Details

```
uname -a
cat /etc/os-release
lsb_release -a
```

8.4 Installed Packages Count

```
dpkg -l | wc -l
```

8.5 Last Login History

```
last
who
```

8.6 Network Speed Test (CLI)

```
sudo apt install speedtest-cli
speedtest-cli
```

9. Auto-run neofetch on Terminal Open

To automatically display system info every time you open a new terminal, add neofetch to your shell startup file.

For bash (default shell):

```
nano ~/.bashrc
# Scroll to the bottom and add:
neofetch
# Save with Ctrl+O, then Enter, then Ctrl+X to exit
# Reload without reopening:
source ~/.bashrc
```

For zsh (if you use oh-my-zsh):

```
nano ~/.zshrc
# Add at the bottom:
neofetch
source ~/.zshrc
```

 NOTE: Replace neofetch with fastfetch if you installed fastfetch — it loads much faster.

Quick Reference Cheat Sheet

Command	What it does
`neofetch`	Full system info with ASCII distro logo
`fastfetch`	Faster modern alternative to neofetch
`df -h`	Disk usage — all filesystems, human readable
`df -h /`	Disk usage — root drive only
`lsblk`	List all drives and partitions
`lsblk -d -o NAME,SIZE,ROTA`	Check SSD (0) vs HDD (1)
`free -h`	RAM total, used, and available
`lscpu`	CPU model, cores, cache info
`top`	Live process monitor (built-in)
`htop`	Better interactive process monitor
`btop`	Best animated system dashboard
`watch -n 1 nvidia-smi`	Live GPU stats (NVIDIA)
`ncdu /`	Interactive disk usage browser
`du -sh /path`	Size of a specific folder
`sensors`	CPU/GPU temperature readings
`uname -a`	Kernel version and architecture
`uptime`	System uptime and load average

— End of Guide —